

Math 142, F20
Exam 2

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	21	
2	24	
3	20	
4	20	
5	21	
Total	106	

Good Luck !

Problem 1. (21 points) Determine whether the **sequence** $\{a_n\}$ is convergent or divergent. If it is convergent, find its limit.

(a) $a_n = \frac{n^2}{e^n}$

let $f(x) = \frac{x^2}{e^x}$, then $a_n = f(n) = \frac{n^2}{e^n}$

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{2x}{e^x} = \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0$$

$$\Rightarrow \left\{ \frac{n^2}{e^n} \right\} \text{ converges and } \lim_{n \rightarrow \infty} \frac{n^2}{e^n} = 0$$

(b) $a_n = \frac{n^3 + 2}{(n^2 + 1)(2n + 1)}$

$$\lim_{n \rightarrow \infty} \frac{n^3 + 2}{(n^2 + 1)(2n + 1)} = \lim_{n \rightarrow \infty} \frac{n^3 + 2}{2n^3 + n^2 + 2n + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{1 + \frac{2}{n^3}}{2 + \frac{1}{n} + \frac{2}{n^2} + \frac{1}{n^3}} = \frac{1}{2}$$

convergent

(c) $a_n = (-1)^{n+1} \cos\left(\frac{1}{n}\right)$

$$\lim_{n \rightarrow \infty} \cos\left(\frac{1}{n}\right) = \cos(0) = 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_{2n-1} = \lim_{n \rightarrow \infty} (-1)^{2n} \cos\left(\frac{1}{2n+1}\right) = 1$$

$$\lim_{n \rightarrow \infty} a_{2n} = \lim_{n \rightarrow \infty} (-1)^{2n+1} \cos\left(\frac{1}{2n}\right) = -1$$

$$\Rightarrow \left\{ (-1)^{n+1} \cos\left(\frac{1}{n}\right) \right\} \text{ diverges}$$

Problem 2. (24 points) Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{+\infty} \frac{2^n}{3^{n+1}}$ (If convergent, please also find the sum for this series)

$$a_n = \frac{2^n}{3^{n+1}} = \frac{1}{3} \left(\frac{2}{3}\right)^n = \frac{2}{9} \left(\frac{2}{3}\right)^{n-1} \Rightarrow \text{Geometric series with } a = \frac{2}{9}, r = \frac{2}{3}$$

$$\frac{2}{3} < 1 \Rightarrow \sum_{n=1}^{+\infty} \frac{2^n}{3^{n+1}} \text{ is convergent.}$$

$$\text{and } \sum_{n=1}^{+\infty} \frac{2^n}{3^{n+1}} = \frac{\frac{2}{9}}{1 - \frac{2}{3}} = \frac{2}{9} \cdot 3 = \frac{2}{3}$$

(b) $\sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^3}$ (Hint: Use the Integration Test)

$$\text{Let } f(x) = \frac{1}{x(\ln x)^3}, \text{ then } a_n = f(n) = \frac{1}{n(\ln n)^3}$$

$$\text{clearly } \frac{1}{n(\ln n)^3} > \frac{1}{(n+1)(\ln(n+1))^3} \text{ when } n > 2.$$

$$\int_2^{+\infty} \frac{1}{x(\ln x)^3} dx = \lim_{b \rightarrow +\infty} \int_2^b \frac{1}{x(\ln x)^3} dx = \lim_{b \rightarrow +\infty} \left[-\frac{1}{2(\ln x)^2} \right]_{x=2}^{x=b}$$

$$= \lim_{b \rightarrow +\infty} -\frac{1}{2(\ln b)^2} + \frac{1}{2(\ln 2)^2} = \frac{1}{2(\ln 2)^2} < +\infty$$

$$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n(\ln n)^3} \text{ is convergent by the Integral Test}$$

(c) $\sum_{n=1}^{+\infty} \frac{\sqrt{4n^2 + 1}}{n^2 + n}$ (Hint: Use the Limit Comparison Test)

$$\text{Let } a_n = \frac{\sqrt{4n^2 + 1}}{n^2 + n}, b_n = \frac{2n}{n^2} = \frac{2}{n^2}$$

$$\lim_{n \rightarrow +\infty} \frac{a_n}{b_n} = 1$$

$$\sum_{n=1}^{+\infty} b_n = 2 \sum_{n=1}^{+\infty} \frac{1}{n^2} \text{ is divergent (by p-series with } p=1)$$

$$\Rightarrow \sum_{n=1}^{+\infty} \frac{\sqrt{4n^2 + 1}}{n^2 + n} \text{ is divergent by the Limit Comparison Test.}$$

Problem 3. (20 points) Determine whether each of the following series is conditionally convergent, absolutely convergent or divergent.

(a) $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$

$u_n = \frac{1}{\ln n}$, clearly $u_n > 0$ when $n \geq 2$.

i) $u_n = \frac{1}{\ln n} > \frac{1}{\ln(n+1)} = u_{n+1}$

ii) $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\ln(n)} = 0$

$\Rightarrow \sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is convergent by the Alternating series Test.

$u_n = \frac{1}{\ln n} > \frac{1}{n}$ when $n \geq 2$ $\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{\ln n}$ is divergent.

Thus $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\ln n}$ is conditionally convergent.

(b) $\sum_{n=1}^{+\infty} \frac{(-1)^{n+1} \sqrt[n]{2}}{n^2}$

$u_n = \frac{\sqrt[n]{2}}{n^2}$

$b_n = \frac{1}{n^2}$

$\lim_{n \rightarrow \infty} \frac{u_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{\sqrt[n]{2}}{n^2}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \sqrt[n]{2} = 1$

$\sum_{n=1}^{+\infty} \frac{1}{n^2}$ is convergent

$\Rightarrow \sum_{n=1}^{+\infty} \frac{\sqrt[n]{2}}{n^2}$ is convergent by the Limit Comparison Test

Thus $\sum_{n=1}^{+\infty} (-1)^{n+1} \frac{\sqrt[n]{2}}{n^2}$ is absolutely convergent.

Problem 4. (20 points) Find the radius of convergence and interval of convergence for the following power series.

(a) $\sum_{n=1}^{+\infty} \frac{(-1)^n x^n}{n 3^n}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{n+1}}{(n+1) 3^{n+1}} \cdot \frac{n 3^n}{(-1)^n x^n} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \cdot \frac{x}{3} \right| = \left| \frac{x}{3} \right| < 1 \Rightarrow |x| < 3$$

$\Rightarrow R=3 \Rightarrow -3 < x < 3$

check two end points: $x=3, -3$

i) $x=3$, $a_n = \frac{(-1)^n 3^n}{n 3^n} = \frac{(-1)^n}{n}$, $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n}$ converges by the Alternating series Test.

ii) $x=-3$, $a_n = \frac{(-1)^n (-3)^n}{n 3^n} = \frac{1}{n}$, $\sum_{n=1}^{+\infty} \frac{1}{n}$ diverges.

thus, the interval of convergence is $[-3, 3]$

(b) $\sum_{n=0}^{+\infty} (n+1)(x-1)^n$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+2)(x-1)^{n+1}}{(n+1)(x-1)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+2}{n+1} \right| |x-1|$$

$$= |x-1| < 1 \Rightarrow R=1$$

$$|x-1| < 1 \Rightarrow -1 < x-1 < 1 \Rightarrow 0 < x < 2$$

check two end points: $x=0, 2$

i) $x=0$, $a_n = (n+1)(-1)^n$, clearly $\sum_{n=1}^{+\infty} (n+1)(-1)^n$ diverges

since $\lim_{n \rightarrow \infty} (n+1)(-1)^n$ DNE.

ii) $x=2$, $a_n = (n+1)1^n = n+1$, clearly $\sum_{n=1}^{+\infty} (n+1)$ diverges

thus, the interval of convergence is $(0, 2)$

Problem 5. (21 Points) Find, and simplify if possible, the Maclaurin series and its radius of convergence for the following functions.

(a) $f(x) = \frac{1}{2}(e^x + e^{-x})$. (Hint: Use $e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!}$ for $|x| < +\infty$)

$$\begin{aligned} f(x) &= \frac{1}{2}(e^x + e^{-x}) = \frac{1}{2} \left(\sum_{n=0}^{+\infty} \frac{x^n}{n!} + \sum_{n=0}^{+\infty} \frac{(-x)^n}{n!} \right) \\ &= \frac{1}{2} \sum_{n=0}^{+\infty} \frac{x^n + (-x)^n}{n!} = \frac{1}{2} \sum_{n=0}^{+\infty} (1 + (-1)^n) \frac{x^n}{n!} \\ &= \frac{1}{2} \sum_{n=0}^{+\infty} \begin{cases} 2 & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases} \frac{x^n}{n!} \\ &= \sum_{n=0}^{+\infty} \frac{x^{2n}}{(2n)!} \quad \text{for } |x| < +\infty \Rightarrow R = +\infty \end{aligned}$$

(b) $f(x) = \frac{x^2}{3-x}$ (Hint: Use $\frac{1}{1-x} = \sum_{n=0}^{+\infty} x^n$ for $|x| < 1$)

$$\begin{aligned} f(x) &= \frac{x^2}{3-x} = \frac{x^2}{3(1-\frac{x}{3})} = \frac{x^2}{3} \frac{1}{(1-\frac{x}{3})} = \frac{x^2}{3} \sum_{n=0}^{+\infty} \left(\frac{x}{3}\right)^n \\ &= \sum_{n=0}^{+\infty} \frac{x^2}{3} \cdot \left(\frac{x}{3}\right)^n = \sum_{n=0}^{+\infty} \frac{x^{n+2}}{3^{n+1}} \\ \text{for } \left|\frac{x}{3}\right| < 1 &\Rightarrow |x| < 3 \Rightarrow R = 3 \end{aligned}$$

(c) $f(x) = (1-x)^{-2}$ (Hint: $(1+x)^m = \sum_{n=0}^{+\infty} \frac{m(m-1)\cdots(m-n+1)}{n!} x^n$ for $|x| < 1$)

$$\begin{aligned} f(x) &= (1-x)^{-2} = [1+(-x)]^{-2} \\ &= 1 + \frac{(-2)}{1!}(-x) + \frac{(-2)(-3)}{2!}(-x)^2 + \frac{(-2)(-3)(-4)}{3!}(-x)^3 + \cdots \\ &\quad + \frac{(-2)(-3)\cdots(-n-1)}{n!}(-x)^n + \cdots \\ &= 1 + 2x + 3x^2 + 4x^3 + \cdots \\ &= \sum_{n=0}^{+\infty} (n+1)x^n \quad \text{for } |x| < 1 \Rightarrow R = 1 \end{aligned}$$